Machine Learning Interview Knowledge Base

Concise answer frameworks for fresher interviews

A practical question bank covering machine learning, model evaluation, deep learning, NLP, recommendation systems, and resume project discussions.

How to use this guide

- Answer in your own words, then add one concrete example. Keep the first answer brief and let the interviewer choose where to go deeper.

- For project questions, replace bracketed prompts with your real data, tools, decisions, and measured results. Do not present sample wording as your actual project history.

- Follow-ups show where an interviewer may probe. A solid answer explains a choice, a trade-off, or a limitation rather than only defining a term.

Questions are grouped by topic. The answer framework is intentionally concise so it can be practiced aloud and personalized.

1. Machine learning fundamentals and algorithms

1. How do supervised and unsupervised learning differ?

Answer framework Supervised learning uses examples with known target answers to learn a mapping from inputs to outputs. Unsupervised learning looks for structure in inputs without target labels.

Fresher example For example, predict whether a customer will buy using past labeled purchases (supervised), or group customers by similar behavior (unsupervised).

Likely follow-ups What is a label? / Can clustering be used before a supervised model?

2. What is the difference between regression and classification?

Answer framework Regression predicts a numeric value; classification predicts a category. The model choice and evaluation metric should match the target.

Fresher example Predicting delivery time in minutes is regression. Predicting whether an order will be returned is classification.

Likely follow-ups Can classification output probabilities? / How would you evaluate each task?

3. Explain linear regression in simple terms.

Answer framework Linear regression estimates a numeric outcome as a weighted combination of input features. It chooses weights that make predictions close to observed values, commonly by minimizing squared error.

Fresher example A simple house-price model might combine area, number of rooms, and age. Its coefficients estimate how each feature relates to price while other features are held fixed.

Likely follow-ups What does a coefficient mean? / What assumptions or limitations should you check?

4. How does logistic regression classify?

Answer framework Logistic regression calculates a weighted score, converts it to a probability with a sigmoid function, and applies a threshold to assign a class. Despite its name, it is commonly used for classification.

Fresher example An order-risk model may estimate a 0.78 chance of cancellation. A threshold such as 0.5 turns that into a predicted cancellation.

Likely follow-ups How would you choose a threshold? / Why might the default threshold be unsuitable?

5. How does a decision tree work, and what is a Random Forest?

Answer framework A decision tree repeatedly splits examples using feature rules to make groups more predictive of the target. A Random Forest averages many trees trained on varied samples and feature subsets, which usually reduces the instability of one tree.

Fresher example A tree might first split customers by prior returns, then by order value. A forest combines many such trees for a more stable prediction.

Likely follow-ups How can a deep tree overfit? / How can you estimate feature importance?

6. When might you use KNN or an SVM?

Answer framework K-nearest neighbors predicts from the labels or values of nearby training examples, so feature scaling and the distance measure matter. An SVM seeks a boundary with a wide margin between classes and can use kernels for nonlinear boundaries.

Fresher example KNN can be a clear baseline for a small, well-scaled dataset. An SVM can work well on a medium-sized dataset with many useful features.

Likely follow-ups What happens as K changes? / Why scale features for these algorithms?

7. What does K-Means do?

Answer framework K-Means partitions numeric examples into K groups by repeatedly assigning each point to its nearest centroid and updating centroids. It needs K in advance and is sensitive to scaling and outliers.

Fresher example A retailer could cluster products using price and sales frequency after standardizing those features.

Likely follow-ups How could you select K? / What shapes of clusters can K-Means struggle with?

2. Model evaluation and improvement

1. Explain accuracy, precision, recall, and F1-score.

Answer framework Accuracy is the fraction of all predictions that are correct. Precision asks what fraction of predicted positives were correct. Recall asks what fraction of actual positives were found. F1 is the harmonic mean of precision and recall.

Fresher example For a rare fraud class, a model can have high accuracy by predicting "not fraud" almost always. Recall and precision reveal whether it catches useful fraud cases.

Likely follow-ups Which metric matters more when missing a positive is costly? / What is a confusion matrix?

2. How do MAE, MSE, RMSE, and R-squared differ?

Answer framework MAE is average absolute error and is easy to interpret in target units. MSE squares errors and penalizes large misses more; RMSE puts that penalty back in target units. R-squared compares model fit with a mean-value baseline, but does not by itself prove a model is good.

Fresher example For delivery-time estimates, MAE of 8 means predictions miss by 8 minutes on average in absolute terms.

Likely follow-ups Which metric is more sensitive to outliers? / Can R-squared be negative on held-out data?

3. What are overfitting and underfitting?

Answer framework Overfitting means a model learns training-specific noise and performs worse on new data. Underfitting means it is too simple or poorly trained to capture useful patterns, so it performs poorly even on training data.

Fresher example A very deep tree can memorize customers in the training set. A straight-line model may underfit a strongly curved pattern.

Likely follow-ups How do training and validation scores help diagnose each case? / Name one remedy for each.

4. Describe the bias-variance trade-off.

Answer framework Bias is error from assumptions that are too simple; variance is sensitivity to the particular training sample. Making a model more complex can reduce bias but increase variance, so validation data helps find a useful balance.

Fresher example A shallow tree may miss real patterns, while a very deep tree may change greatly when the training examples change.

Likely follow-ups How do regularization and more data affect this trade-off?

5. What is cross-validation, and how do you avoid leakage?

Answer framework In K-fold cross-validation, data is divided into K parts; each part is held out once while the model trains on the others. Fit preprocessing only on each training fold, and keep a final test set separate until choices are complete.

Fresher example Scale and impute inside a pipeline so validation-fold statistics do not leak into training.

Likely follow-ups How would you split time-ordered data? / What is target leakage?

6. What is feature engineering?

Answer framework Feature engineering creates or transforms inputs so that relevant patterns are easier for a model to learn. It should use information available at prediction time and be evaluated on held-out data.

Fresher example For a purchase prediction, useful features might include days since last order, average basket value, or product category counts.

Likely follow-ups How do you encode a category? / How would you check that a feature is available at inference time?

3. Deep learning

1. What is a neural network?

Answer framework A neural network is a set of connected layers that transform inputs through learned weights and biases. During training, backpropagation computes how weights contributed to the loss, and an optimizer updates them.

Fresher example A small network could combine numeric customer features to predict the chance of a repeat purchase.

Likely follow-ups What are weights and biases? / What does backpropagation calculate?

2. Why do neural networks use activation functions?

Answer framework Activation functions add nonlinear behavior. Without them, stacking layers would still be equivalent to one linear transformation and could not model many curved relationships.

Fresher example ReLU is commonly used in hidden layers; sigmoid is often used to represent a binary-class probability.

Likely follow-ups What can happen with sigmoid in deep hidden layers? / Why is softmax used for multiple classes?

3. What is a loss function?

Answer framework A loss function measures the mismatch between model predictions and target answers. Training aims to reduce this value; the chosen loss should match the task.

Fresher example Mean squared error is common for numeric prediction; cross-entropy is common for classification.

Likely follow-ups How is loss different from accuracy? / What loss would suit a regression task?

4. What are epochs and batch size?

Answer framework An epoch is one pass through the training examples. Batch size is how many examples contribute to one weight update. These choices affect training speed, memory use, and the noisiness of updates.

Fresher example With 1,000 examples and batch size 100, one epoch has 10 batches.

Likely follow-ups What is an iteration? / What might you observe if the learning rate is too high?

5. When are CNNs and RNNs useful?

Answer framework Convolutional neural networks use filters to learn local spatial patterns and are widely used for images. Recurrent neural networks process sequences while carrying information across steps, which can suit sequential data such as text or sensor readings.

Fresher example A CNN can classify product photos; an RNN can process a short sequence of words. Transformer models are also widely used for language tasks.

Likely follow-ups What does a convolution filter learn? / Why can long sequences be difficult for a basic RNN?

4. Natural language processing

1. What are common text preprocessing steps?

Answer framework Depending on the task, preprocessing can include cleaning markup, normalizing case, handling punctuation, tokenizing, and deciding how to treat stop words or rare terms. Keep steps consistent between training and inference.

Fresher example For a short review classifier, remove HTML, tokenize the text, and preserve negation: "not good" should not become just "good".

Likely follow-ups When might lowercasing hurt? / Why can removing stop words change meaning?

2. What is tokenization?

Answer framework Tokenization splits text into units such as words, subwords, or characters. Modern language models often use subword tokens so that uncommon words can be represented as combinations of familiar pieces.

Fresher example A word like "unhelpful" could be represented as several subword pieces.

Likely follow-ups Why do different tokenizers produce different token counts? / What is an out-of-vocabulary word?

3. Explain TF-IDF.

Answer framework TF-IDF gives a term more weight when it appears often in a document but is uncommon across the collection. It turns text into numeric features useful for simple search and classification baselines.

Fresher example The word "excellent" may be informative in reviews if it appears in some reviews but not almost all of them.

Likely follow-ups How is TF-IDF different from an embedding? / What role does document frequency play?

4. How would you build sentiment or emotion classification?

Answer framework Define the label set, gather representative labeled examples, split data without leakage, build a simple baseline such as TF-IDF plus logistic regression, and evaluate per-class precision and recall. Inspect errors and label quality before trying more complex models.

Fresher example Sentiment might use positive, neutral, and negative labels; emotion could use joy, sadness, anger, and fear, with more than one label per text if appropriate.

Likely follow-ups How do sentiment and emotion differ? / How would you handle class imbalance or ambiguous labels?

5. Recommendation systems

1. What is collaborative filtering?

Answer framework Collaborative filtering recommends items using patterns in user-item interactions. It can use similar users or learn user and item representations from interaction data, without needing rich item descriptions.

Fresher example If users who bought the same books also liked a particular title, that title may be recommended to another similar user.

Likely follow-ups How is this different from content-based filtering? / What data represents an interaction?

2. What is user-based recommendation?

Answer framework User-based recommendation finds users with interaction histories similar to the target user, then suggests items those neighbors liked that the target user has not seen.

Fresher example If two readers rate many of the same books similarly, recommend a book one liked that the other has not rated.

Likely follow-ups How might similarity be measured? / What happens when a new user has no history?

3. What is sparsity in recommendation data?

Answer framework The user-item matrix is sparse because each user interacts with only a small fraction of all available items. Sparse data makes similarity estimates less reliable and worsens cold-start problems.

Fresher example A new book with only one rating is hard to place confidently based on collaborative patterns alone.

Likely follow-ups How can side information help? / What is the cold-start problem?

4. What does HitRate@K measure?

Answer framework HitRate@K is the fraction of test cases where at least one relevant item appears among the top K recommendations. It is easy to interpret, but ignores the position of the hit and the number of relevant items retrieved.

Fresher example If a relevant book appears in the top 5 for 7 of 10 users, HitRate@5 is 0.7.

Likely follow-ups How does it differ from precision@K or NDCG? / How would you create a fair offline test split?

6. Resume project questions

1. Walk me through your e-commerce prediction project.

Answer framework Use a clear sequence: business question, prediction target, data and split, baseline, model and metric, result, and limitation. State only the tools and results you actually used; distinguish your own contribution from team work.

Fresher example Example framework: "I predicted [target] from [available-at-decision-time features]. I compared [baseline] with [model], used [metric] because [reason], and found [measured result]. The main limitation was [limitation]."

Likely follow-ups Why did you choose that target? / What was your baseline? / How would the business use a prediction?

2. How did you prevent data leakage in the e-commerce project?

Answer framework Explain the time at which a prediction would be made, then show that every feature and split respects that time. Fit preprocessing on training data only and use a time-based split if future behavior is being predicted.

Fresher example For a return prediction made at checkout, do not use return status or events recorded after checkout as features.

Likely follow-ups Which feature was most at risk of leakage? / How would you validate on a later time period?

3. How does your book recommendation project work?

Answer framework Describe the interaction data, recommendation approach, candidate generation, ranking or scoring, and evaluation. If it is collaborative filtering, explain how user-item patterns lead to unseen-book suggestions; be clear about cold-start behavior.

Fresher example A simple fresher project could recommend books liked by similar readers and exclude books the current reader already rated.

Likely follow-ups How did you measure recommendation quality? / What happens for a new reader or new book?

4. How did you handle sparse feedback in the book project?

Answer framework Explain what counts as feedback, how few interactions affect similarity or ranking, and any fallback such as popular books or book metadata. Avoid claiming strong personalization for users with little history.

Fresher example For a new reader, show popular or genre-matched books until enough interactions are available.

Likely follow-ups Did you compare against a popularity baseline? / How might implicit feedback differ from explicit ratings?

5. How did you build the emotion classification project?

Answer framework State the emotion labels, source and size of labeled data, preprocessing, model baseline, evaluation, and how you inspected mistakes. Mention whether examples could have multiple emotions and how class imbalance was handled.

Fresher example A concise example: "I cleaned and tokenized text, trained a TF-IDF classifier, and reviewed per-class recall because missing a minority emotion matters." Adapt this to the actual project.

Likely follow-ups How did you define the labels? / Which emotions were confused most often? / How would you improve it?

6. What was a challenge across your projects, and what would you improve?

Answer framework Choose a real challenge and describe your action and evidence: problem, options considered, what you changed, and what you learned. Finish with one practical next step rather than claiming the project is perfect.

Fresher example For example, limited labels may motivate collecting more representative examples, checking annotation consistency, or improving validation before changing the model.

Likely follow-ups What did you personally own? / What would you do first with more time? / What did not work as expected?

Quick practice checklist

- Can I explain the problem and target in one sentence?
- Can I justify my baseline, model choice, split, and evaluation metric?
- Can I give one concrete example and one limitation without overstating results?
- For resume projects, can I state my contribution and next improvement?
- If I do not know an answer, can I reason from fundamentals and say how I would verify it?

Project answer template

> *Problem and target > data and split > baseline > model choice > metric and result > limitation and next step.*